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The oxidation of *n*-butylbenzene: Experimental study in a JSR at 10 atm and detailed chemical kinetic modeling

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Abstract

The oxidation of *n*-butylbenzene was studied in a jet-stirred reactor (JSR) at 10 atm in dilute conditions providing new experimental results over the low- and high-temperature range 550–1180 K, and variable equivalence ratio ($0.25 \leqslant \varphi \leqslant 1.5$). They consisted of concentration profiles of the reactants, stable intermediates and final products, measured as a function of temperature, at a constant residence time of 1 s, by sonic probe sampling followed by on-line GC–MS and off-line GC–TCD–FID and GC–MS analyses. The oxidation of *n*-butylbenzene in these conditions was modeled using a detailed chemical kinetic reaction mechanism (404 species and 2210 reactions, most of them reversible) deriving from a previous scheme proposed for the ignition, oxidation, and combustion of simple aromatics (benzene, toluene, styrene, *n*-propylbenzene). Sensitivity analyses and reaction path analyses, based on rates of reaction, were used to interpret the results. The proposed kinetic scheme could also be used to represent literature data (plug-flow reactor oxidation of *n*-butylbenzene, ignition delays in a rapid compression machine (RCM), a low-pressure methane–oxygen–argon flame doped with *n*-butylbenzene), to confirm its validity.
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1. Introduction

Aromatic hydrocarbons represent an important fraction of commercial gasoline (ca. 14% in wt.), Diesel fuel (ca. 25% in wt.), and kerosene (ca. 25% in vol.) [1]. Nevertheless, their oxidation and pyrolysis are still not fully understood whereas, in order to improve engines efficiency and reduce the emission of pollutants and toxics, detailed modeling of combustion in engines is frequently used. Since the composition of commercial fuels is very complex, combustion kinetic schemes for simple surrogate fuels are needed [2–5]. Whereas the kinetics of oxidation of light aromatics has already been investigated extensively, only few studies concern the oxidation of larger alkylbenzenes such as those found in kerosene and Diesel fuels. Furthermore, up to now, the low-temperature oxidation of alkylbenzenes, relevant to Diesel combustion, has only been studied in a RCM [6,7] over the temperature range 640–840 K and the formation of several cyclic ethers resulting from the low-temperature oxidation of the fuel's alkyl-chain were reported. Their results were used to propose a low-temperature oxidation mechanism [6]. Brezinsky et al. [8] studied

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the high-temperature oxidation of *n*-butylbenzene and proposed a reaction scheme based on their flow reactor experiments performed at 1069 K and for an equivalence ratio of 0.98. Anderson et al. [9] studied the combustion of a non-premixed methane flame seeded with ca 500 ppm of *n*-butylbenzene at 1atm. More recently, Pousse et al. [9] studied the oxidation of *n*-butylbenzene in a low-pressure premixed methane–argon–oxygen flame. Their flame structures were used together with the results of Brezinsky et al. [8] to propose a high-temperature kinetic scheme.

More information was needed to validate a chemical kinetic scheme covering both the low- and high-temperatures regimes. Therefore, as part of continuing efforts for better understand the oxidation kinetics of hydrocarbons, the oxidation of *n*-butylbenzene was investigated experimentally over a wide range of conditions using a JSR and the results were interpreted in terms of a detailed kinetic mechanism also used to simulate literature experiments.

2. Experimental

The JSR experiment is similar to that described earlier [11,12]. The reactor consisted of a sphere of 40 mm diameter (33 cm^3) made of fused silica (to minimize wall catalytic reactions), equipped with 4 nozzles of 1 mm I.D. for the admission of the gases which achieve the stirring. A nitrogen flow of 100 L/h was used to dilute the fuel and avoid its pyrolysis before admission into the reactor. The gases were preheated before injection to minimize temperature gradients inside the JSR. The reactants were diluted by nitrogen and mixed at the entrance of the injectors. High purity reactants were used: oxygen was 99.995% pure and *n*-butylbenzene (CAS 104-51-8) > 99% pure. Furthermore, *n*-butylbenzene was sonically degassed before use. A piston pump (Shimadzu LC-10AD VP) was used to deliver the fuel to an atomizer–vaporizer assembly maintained at 200 °C. A good thermal homogeneity along the whole vertical axis of the reactor was observed for each experiment by thermocouple (0.1 mm Pt–Pt/Rh 10% located inside a thin-wall silica tube) measurements (temperature change $\leqslant$5 K). The reacting mixtures were sampled via a low-pressure fused-silica sonic probe. The samples (4 kPa) were taken at steady temperature and residence time. They were analyzed on-line by a GC–MS or FID and off-line after collection and storage in 1 L Pyrex bulbs. Low vapor-pressure compounds were analyzed on-line whereas permanent gases and high vapor-pressure species were analyzed off-line. All the products were analyzed by a chromatographic peak identified. The experiments were performed at steady state, at a constant mean residence time of 1 s, the reactants flowing continually in the reactor, varying stepwise the temperature of the gases inside the JSR. A high degree of dilution (0.1% vol. of fuel) was used, reducing temperature gradients in the JSR and heat release (no flame occurred in the JSR). Several gas chromatographs (GC), equipped with capillary columns (Poraplot-U, Molecular Sieve-5A, DB-5ms, DB-624, Plot Al_2O_3/KCl, Carboplot-P7), thermal conductivity detector (TCD) and flame ionization detector (FID), were used for stable species measurements. Compound identifications were made through GC/MS (Varian Saturn 2000) analyses of the samples. CH_2O and CO_2 were measured by FID after hydrogenation on a methanizer (H_2/Ni catalyst) at the exit of the Poraplot-U GC column. The uncertainties associated with the GC measurements were ca 10%. A good repeatability of the measurements and a reasonably good carbon balance (100 ± 10%) were obtained in this series of experiments.

3. Modeling

The JSR modeling was performed using the PSR computer code [13] that computes species concentrations from the balance between the net rate of production of each species by chemical reaction and the difference between the input and output flow rates of species. The kinetic reaction mechanism used here is based on a comprehensive mechanism developed for the oxidation of natural gas blends to diesel fuel [12] with a sub-mechanism for *n*-butylbenzene oxidation deriving from those proposed earlier for *n*-propylbenzene [3,14] and for *n*-butylbenzene by Ribaucour et al. [6]. The reaction mechanism used here consisted of 2210 reversible reactions involving 404 species. It has a strong hierarchical structure. Since most of the present kinetic reaction mechanism has been presented in detail in previous papers [3,5,12,14,15], only the reaction sub-mechanism for the oxidation of *n*-butylbenzene is presented. The aromatic oxidation sub-scheme previously used [3] was updated. This concerns the reaction of the benzyl radical with HO_2, $C_6H_5CH_2 + HO_2 = C_6H_5CH_2O + OH$, reactions of the benzene sub-scheme ($C_6H_6 + HO_2 = C_6H_5 + H_2O_2$ [16], $C_6H_6 + OH = C_6H_5 + H_2O$ [17], $C_6H_5 + O_2 = C_6H_5O + O$, based on Emdee [18], $C_6H_5O = C_5H_5 + CO$ [19], reactions of toluene ($C_6H_5CH_3 + OH = C_6H_5CH_2 + H_2O$ [19], $C_6H_5CH_3 + OH = C_6H_5OH + CH_3$ [20], $C_6H_5CH_3 + H = C_6H_5CH_2 + H_2$ [21]), reactions of ethylbenzene ($C_6H_5C_2H_5 = C_6H_5CH_2 + CH_3$, $C_6H_5C_2H_5 = C_6H_5 + C_2H_5$, $C_6H_5C_2H_5 = H + C_6H_5CH_2CH_2$ [22]), and reactions of the propylbenzene sub-scheme ($C_6H_5CH_2CH_2CH_2 = C_2H_4 + C_6H_5CH_2$, $C_6H_5CH_2CHCH_3 = C_6H_5 + C_3H_6$, [22]). The rate constant for $C_6H_5CH_2CH_2CH_3 + H = nC_3H_7 + C_6H_6$ was estimated based on the corresponding reaction for toluene. For several low-temperature reactions of *n*-butylbenzene, the kinetics of Ribaucour et al. [6] was updated using recent results of DeSain et al. [23] ($C_6H_5(CH_2)_3CH_2 +$

$O_2 = C_6H_5(CH_2)_3CH_2O_2$, $C_6H_5(CH_2)_2CHCH_3 + O_2 = C_6H_5(CH_2)_2CH(O_2)$ CH_3,$C_6H_5CH_2CHCH_2$-$CH_3 + O_2 = C_6H_5CH_2CH$ $(O_2)CH_2CH_3)$ and Ref. [24] ($C_6H_5CH(CH_2)_2CH_3 + O_2 = C_6H_5CH$-$(O_2)(CH_2)_2CH_3$). The rate of decomposition reactions of ketohydroperoxides were decreased ($k = 1.5.10^{16}\ \exp(-22393/T)$), and the metathesis reactions of cyclic ethers with H, OH and HO_2 were modified and the products issued from the cyclic ether radical were modified according to the structure of the parent cyclic ether. A high-temperature sub-mechanism for *n*-butylbenzene was also added according to the scheme proposed by Brezinsky [8]: unimolecular initiation reactions, decomposition of phenylbutyl radical by C—C and C—H bond breaking and isomerization reactions of phenylbutyl radicals. The rates of reaction were computed from the kinetic reaction mechanism and the rate constants of the elementary reactions calculated at the experimental temperature, using the modified Arrhenius equation, $k = A \times T^b \times \exp(-E/RT)$. The rate constants for the reverse reactions were computed from the forward rate constants and the equilibrium constants ($K_c = k_{forward}/k_{reverse}$) calculated using the appropriate thermochemical data [25–27]. The pressure dependencies of the unimolecular reactions and of some pressure-dependent bimolecular reactions were taken into account when information was available (i.e., $k(P,T)$). In the proposed *n*-butylbenzene sub-mechanism the formation reactions of the cyclic ethers and the decomposition reactions of both ketohydroperoxides and cyclic ethers are written without reverse rates. Moreover, none of the reactions of this sub-mechanism exhibits pressure dependence. Local first-order sensitivity analyses and reaction rates analyses were performed computing rates of consumption (ROC) and production (ROP) for every species.

4. Results and discussion

4.1 JSR conditions

The main species measured were the fuel, oxygen, carbon monoxide, carbon dioxide, hydrogen, methane, ethylene, propene, benzene, styrene, formaldehyde, acetaldehyde, and benzaldehyde. Toluene, ethylbenzene, acetylene, 1,3-butadiene, propyne, oxiran, methyloxiran, acetone, and 2-propenylbenzene were also measured at lower concentrations. The products detected here and their relative abundances are in line with the experimental results obtained in a previous plug-flow reactor study [8]. In RCM [8], the authors used much higher fuel concentrations than in this work. Therefore, we could not quantify low-temperature oxidation species reported previously in RCM (cyclic ethers). However, the following oxygenated products were measured under JSR conditions: 1-phenyl-1-butanone, 1-phenyl-2-butanone.

Figs. 1–4 compare the modeling results obtained for the oxidation of *n*-butylbenzene in a JSR, using the presently proposed scheme. The present kinetic scheme was also used to successfully simulate the oxidation of benzene, toluene, and *n*-propylbenzene in similar conditions [12,28,14]. As can be seen from these figures, the model represents fairly well the present data. However, the model underestimates the concentrations of CO and CO_2 at high-temperature (>800 K). Also, the model underestimates acetaldehyde and benzene in fuel-lean conditions. Propene concentrations are overestimated at low-temperature. The model predicts a transition to high-temperature at temperatures slightly too low which results in an overestimation of the concentrations of intermediates in the temperature range 700–780 K. Finally, the model over predicts the formation of CO and aldehydes at low-temperature.

Sensitivity analyses and reaction path analyses, based on species reaction rate of production and rate of consumption computations, were used to interpret the results (see Supplementary material). According to these computations in the present conditions, at low-temperature (650 K), the oxidation of *n*-butylbenzene occurs via metathesis reactions with OH yielding four phenylbutyl radicals. These radicals mostly react by addition with oxygen yielding phenylbutylperoxy radicals whereas the formation of phenylbutenes represents a minor route. Phenylbutylperoxy radicals isomerize yielding hydroperoxyphenylbutyl radicals (QO_2H). Due to their higher stability, the benzylic isomers dominate. QO_2H radicals decompose to form phenylbutenes and cyclic ethers or react by addition with O_2 forming O_2QO_2H radicals. In turn, these radicals isomerize into $HO_2Q'O_2H$ that produce carbonyl compounds and OH radicals via decomposition. At 850 K, butylbenzene still reacts through metathesis yielding 4 phenylbutyl radicals. The radical 1-phenyl-4-butyl (AC_6H_5-C_4H_8) mostly isomerizes (92%) into 1-phenyl-1-butyl ($DC_6H_5C_4H_8$); this represents 50% of its formation route. This benzylic radical decomposes (45%) and reacts by addition with oxygen (55%) forming RO_2 radicals. The RO_2 radicals isomerize into QO_2H that in turn decompose forming cyclic en ethers, aldehydes, ketones and/or alkenes. The other two phenylbutyl radicals ($BC_6H_5C_4H_8$ and $CC_6H_5C_4H_8$) are oxidized forming phenylbutenes (64% for $BC_6H_5C_4H_8$ and 43% for $CC_6H_5C_4H_8$), and decompose via β-scission C—C. At 1050 K, the reaction paths are similar to those delineated at 850 K. However, the relative importance of the reaction routes is different. Butylbenzene still reacts through metathesis and the 1-phenyl-4-butyl radical is still consumed by isomerization into 1-phenyl-1-butyl (94%) which in turn decomposes by β-scission C—C yielding styrene. The two other phenylbutyl radicals decompose (69% for 1-phenyl-3-butyl and 83% for phenyl-2-butyl) or

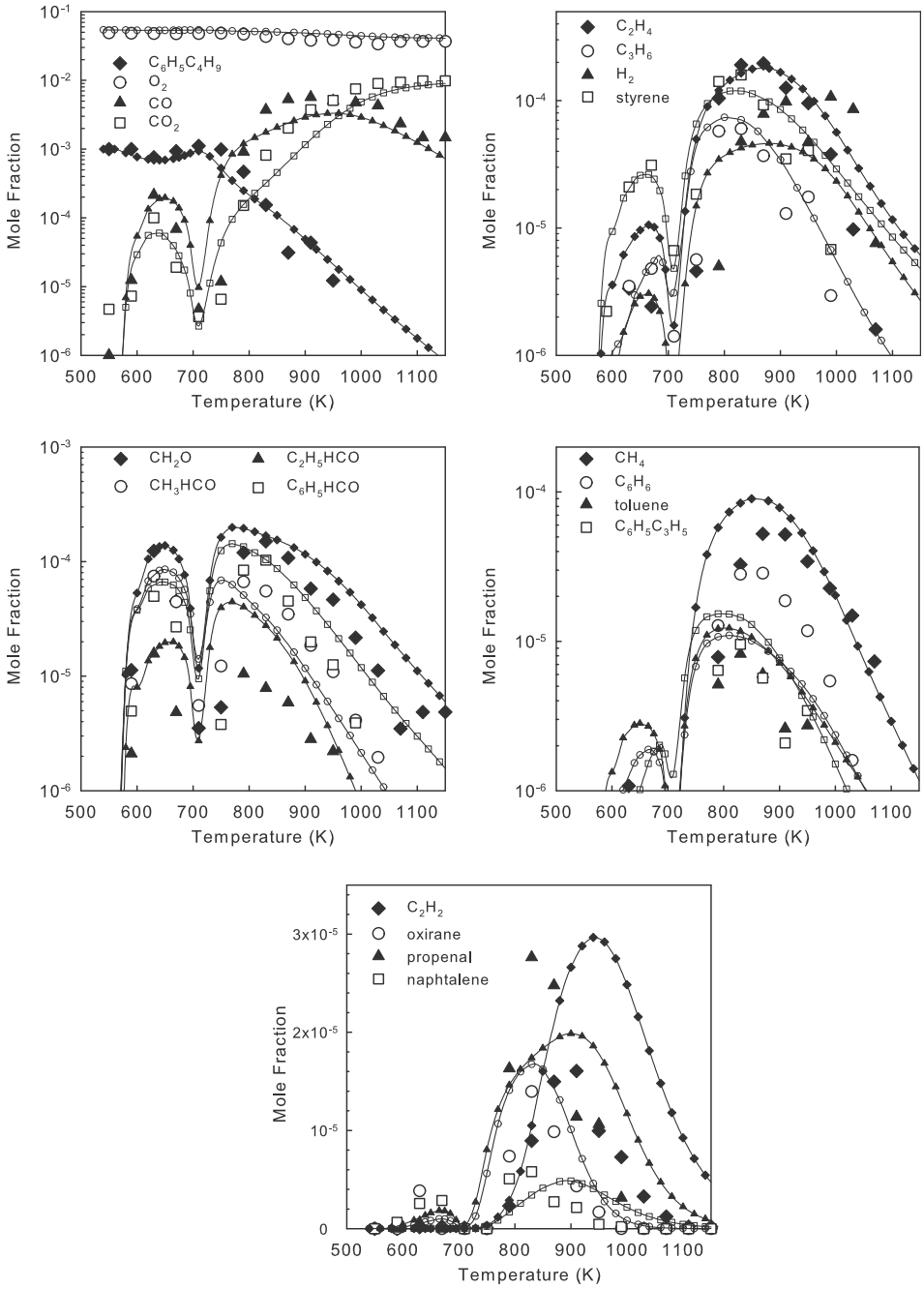

Fig. 1. The oxidation of 1000 ppm of *n*-butylbenzene in a JSR at 10 atm ($\varphi = 0.25$, 1 s). The data (large symbols) are compared to the computations (lines and small symbols).

oxidize (30% for 1-phenyl-3-butyl and 10% for 1-phenyl-2-butyl).

Sensitivity analyses performed for the $\varphi = 0.25$ mixture indicated that the kinetics of *n*-butylbenzene + OH reactions have a strong influence on the computations. Below 600 K, these reactions favor the consumption of *n*-butylbenzene. Between 600 and 720 K the sensitivity coefficients for the reactions $C_6H_5C_4H_9 + OH = bC_6H_5C_4H_8 + H_2O$ and $C_6H_5C_4H_9 + OH = cC_6H_5C_4H_8 + H_2O$ increase to positive values indicating these reactions do not anymore favor the overall oxidation

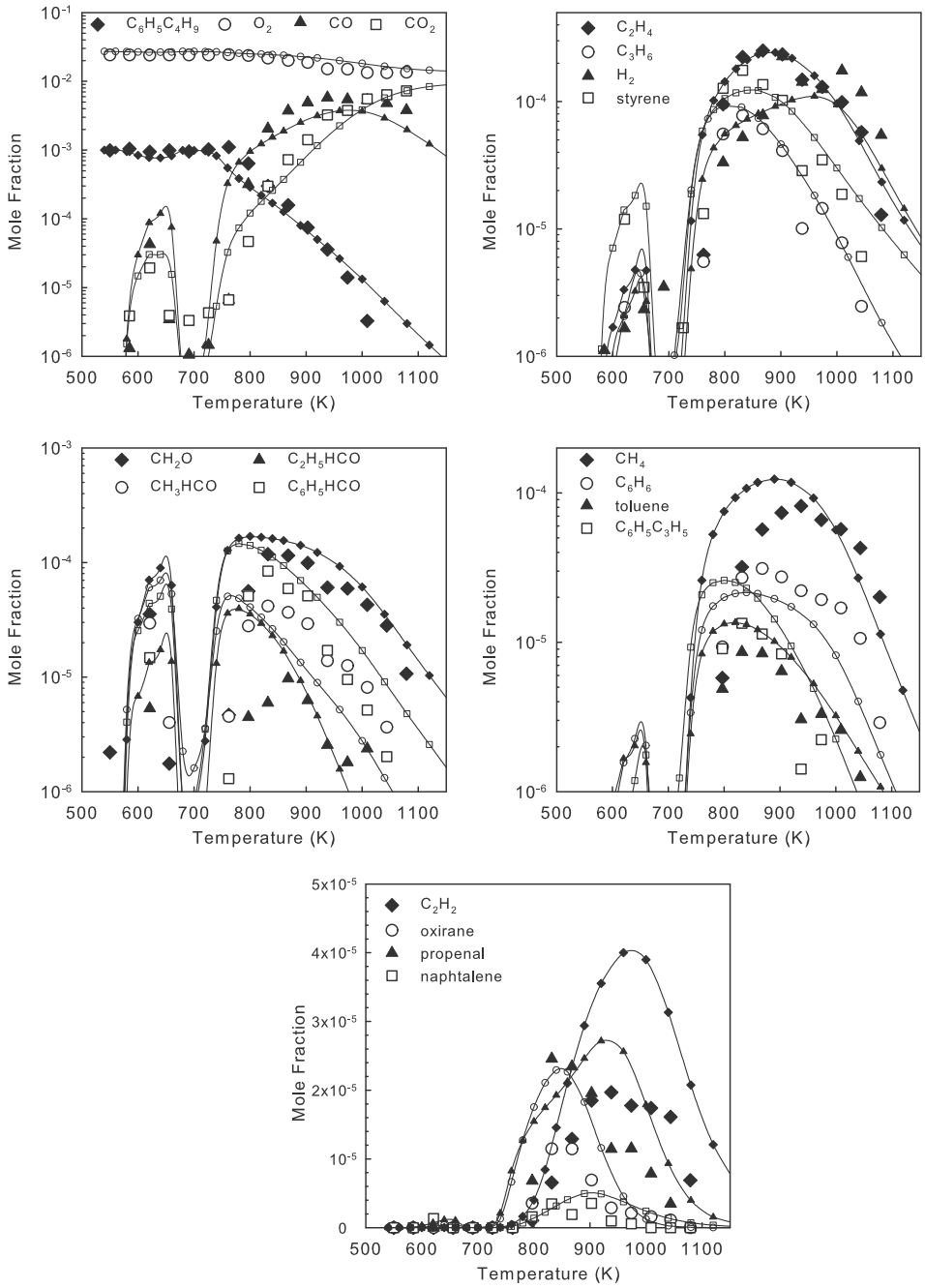

Fig. 2. The oxidation of 1000 ppm of *n*-butylbenzene in a JSR at 10atm ($\varphi = 0.5$, 1 s).

of the fuel. This temperature window corresponds to a transition in the mechanism where phenylbutyl radical peroxidation is replaced gradually by oxidation processes yielding phenylbutenes. At higher temperatures, the influence of the oxidation routes declines in favor of decomposition paths involving β-scission. There, the sensitivity coefficients of metathesis reactions diminish. Similar trends are observed for the other reaction of *n*-butylbenzene with OH. The consumption of *n*-butylbenzene also depends on the kinetics of the phenoxy radical: $C_6H_5O = C_5H_5 + CO$ and $C_6H_5O + H\,(+M) = C_6H_5OH\,(+M)$, particularly between 800 and 950 K.

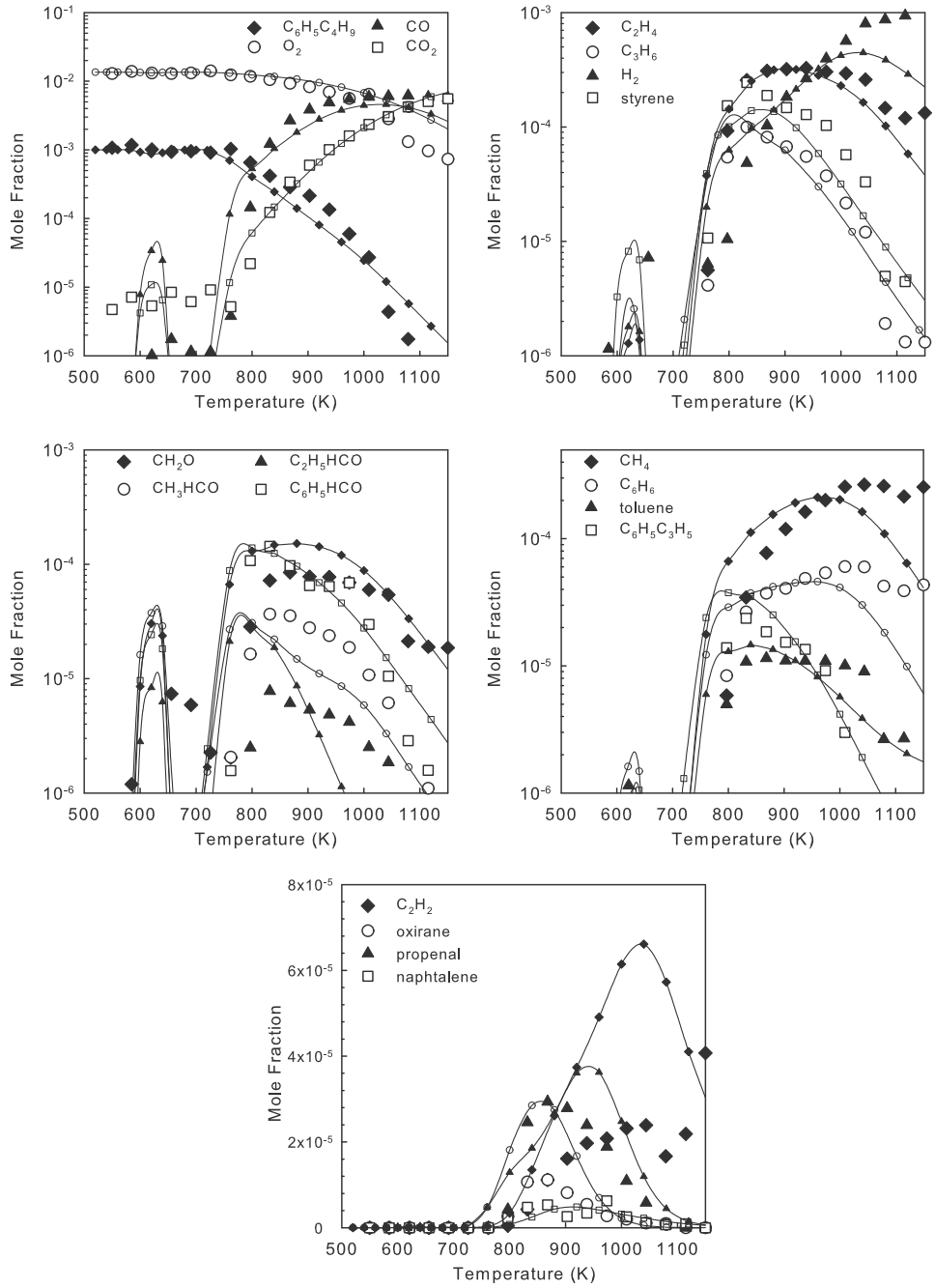

Fig. 3. The oxidation of 1000 ppm of *n*-butylbenzene in a JSR at 10 atm ($\varphi = 1$, 1 s).

4.2 Other conditions

To further assess the validity of the proposed kinetic scheme, the ignition of *n*-butylbenzene–air mixtures in a RCM [6,7] not simulated previously was modeled using the Senkin computer code [29]. As can be seen from Fig. 5, the model predicts well the RCM data. Since an uncertainty of 30 K was reported for the temperature determination in the RCM, computations were performed at $T \pm 15$ K. The results are presented in Fig. 5 as error bars. Sensitive reactions for RCM were determined to be the oxidation reactions of phenylbutyl radicals to form phen-

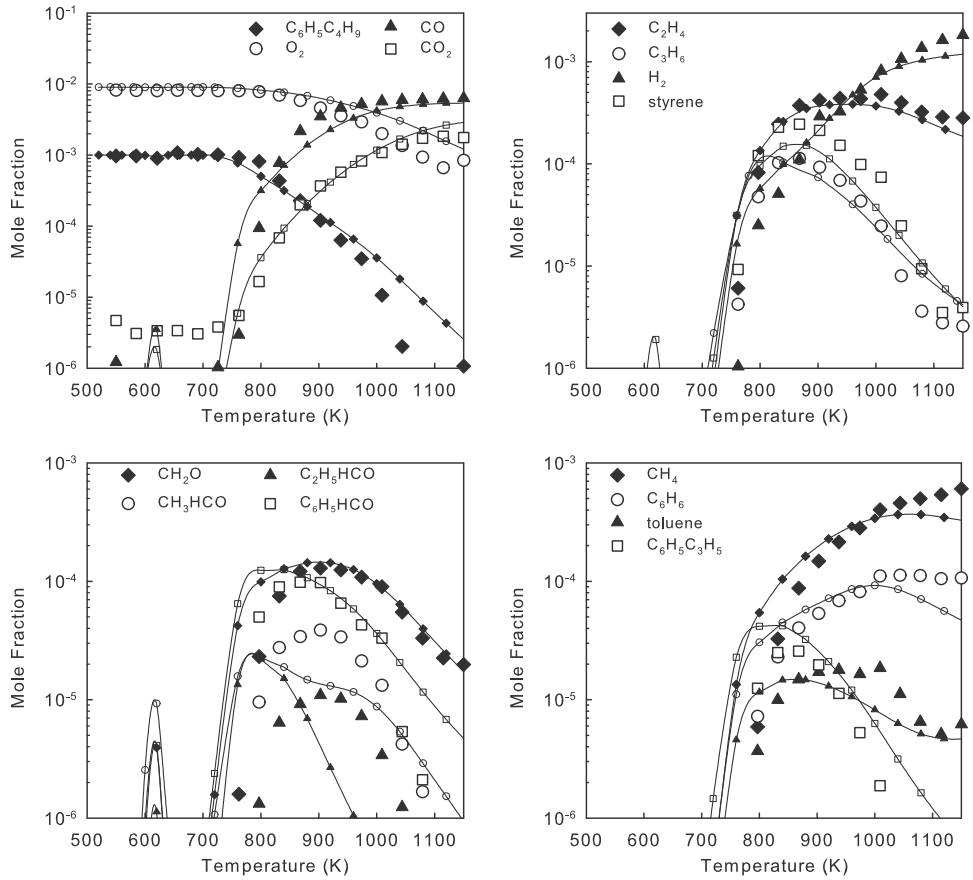

Fig. 4. The oxidation of 1000 ppm of *n*-butylbenzene in a JSR at 10 atm ($\varphi = 1.5$, 1 s).

ylbutene ($C_6H_5C_4H_8 + O_2 = C_6H_5C_4H_7 + HO_2$), the metathesis reactions of *n*-butylbenzene with OH and HO_2 to form the 1-phenyl-1-butyl radical ($C_6H_5C_4H_9 + OH = dC_6H_5C_4H_8 + H_2O$ and $C_6H_5C_4H_9 + HO_2 = dC_6H_5C_4H_8 + H_2O_2$), the reaction between hydroperoxy radicals HO_2 to form hydrogen peroxide ($HO_2 + HO_2 = H_2O_2 + O_2$). For the lowest temperature the ignition delay is also sensitive to the decomposition reactions of ketohydroperoxides, and to the decomposition reaction of H_2O_2 to give 2 hydroxyl radicals ($OH + OH\,(+M) = H_2O_2\,(+M)$) for temperature higher than 750 K.

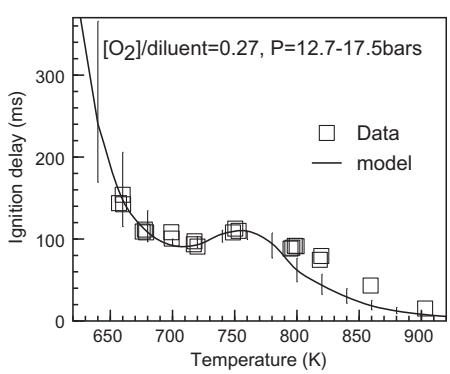

Fig. 5. The ignition of *n*-butylbenzene in a RCM ($\varphi = 1$, mix c, [7]).

Although not shown here, it should be stated that this model was also tested against the experiments performed in a turbulent-flow reactor [8] and a low-pressure (6.7 kPa) methane–oxygen–argon flame seeded with *n*-butylbenzene [10]. The simulations were performed using respectively the Senkin and Premix computer codes [29,30]. Under flow-reactor conditions, the modeling results were in reasonable agreement with the data for the main species: *n*-butylbenzene, styrene, and ethylene whereas the concentrations of oxygenated species were underestimated (phenol, benzaldehyde, and CO). The under-prediction of ethylbenzene could be due to the formation of this product by recombination reactions during the sampling, as for dibenzyl. The model overestimated the formation of propene, benzene, acetylene, and propenylbenzene.

Under low-pressure flame conditions the simulations were in good agreement with the data (Supplementary data).

5. Conclusion

New experimental results were obtained in a JSR for the kinetics of oxidation in dilute conditions of *n*-butylbenzene at 10 atm, for temperatures in the range 550–1180 K, and equivalence ratios of $0.25 \leqslant \varphi \leqslant 1.5$. Concentration profiles of the reactants, stable intermediates and final products were measured by sonic probe sampling followed by on-line GC–MS and off-line GC–TCD–FID–MS analyses. A detailed chemical kinetic reaction mechanism derived from a previous scheme was proposed to model the oxidation of *n*-butylbenzene in these conditions. Sensitivity analyses and reaction path analyses, based on rates of reaction, were used to interpret the results. It could also be used to simulate literature experiments in a RCM, a plug-flow reactor, and a low-pressure premixed flame.

Appendix A. Supplementary data

This publication includes the following supplementary data: the proposed chemical kinetic reaction mechanism in CHEMKIN format, the associated thermochemical data in CHEMKIN format and other modeling tests. Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.013.

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